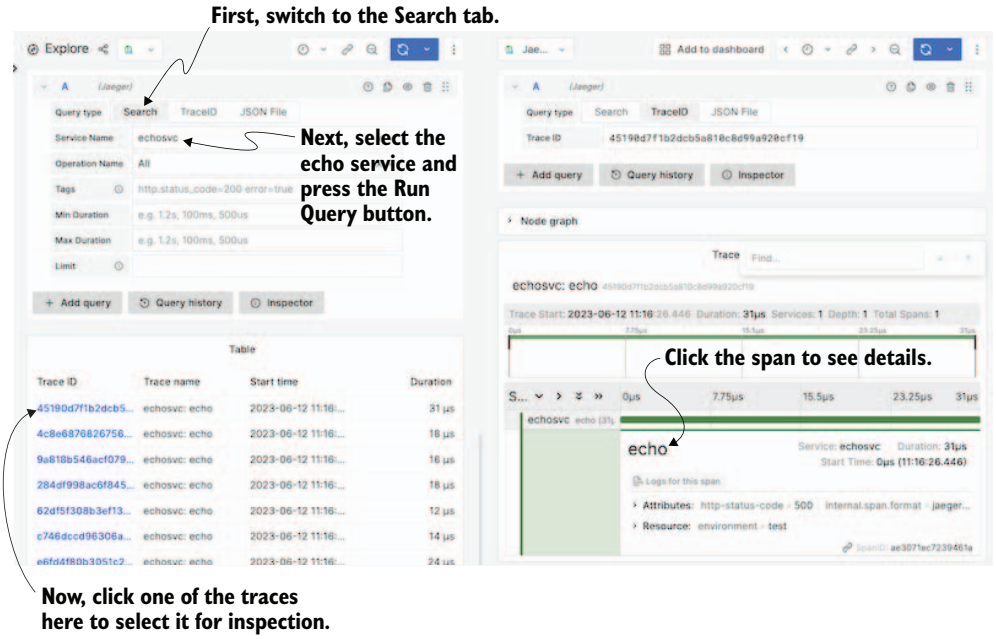


Figure 2.6 Exploring traces with Jaeger: detailed trace view

2.5.3 Costs and benefits

Let's take a closer look at the costs and benefits of the traces signal type. Table 2.3 provides a cost overview of traces.

Table 2.3 Overview of the costs of traces

	Overhead	Considerations
Instrumentation	Some effort and auto-instrumentation	OpenTelemetry uptake and legacy
Agent	Sampling strategy and latency overhead	Use OpenTelemetry collector
Destination	Dedicated storage layer and query and analytics	Lots of commercial offerings and some OSS (e.g., Tempo and Jaeger)

Instrumentation costs of traces

The instrumentation costs of traces, from a developer's point of view, look as follows: the original code in listing 2.1 has 24 lines of code, and the code instrumented with traces in listing 2.10 has 99 lines of code overall.

Now, to isolate the traces-caused instrumentation costs from the total (logs + metrics + traces) instrumentation cost, we need to subtract the 8 LOC from the logs-caused instrumentation as well as the 24 LOC that the metrics instrumentation contributed.